

ULAGANATHAN A

QA Automation Engineer • SDET • Full Stack Developer (MERN)

Chennai / Bengaluru / Remote • ulaganathana7@gmail.com • 8300601336 • Immediate Joiner

3+ Years

QA Automation
Healthcare & Fintech

40%

S3 Validation Reduction
Automation • Fintech

Philips

Key Contributor
Rel26H2 Reliability

PROFESSIONAL SUMMARY

Results-driven QA Automation Engineer & Full Stack Developer with **3+ years of industry experience** across Healthcare (Philips) and Fintech (Banking & Payments) domains. Expert in Python, Behave BDD, SpecFlow, Selenium, and API automation with deep expertise in OCR validation, OpenCV computer vision, and CI/CD pipeline integration. Delivered clinical-grade quality assurance for Philips IVUS imaging under IEC 62304 standards. MERN-certified full stack developer. Currently expanding into Playwright for modern cross-browser automation.

TECHNICAL SKILLS

Languages: Python • JavaScript • C# • Java (Basic)

Automation Frameworks: Behave (BDD) • SpecFlow (BDD) • Selenium WebDriver • Playwright (Learning) • Rest Assured • Cucumber / TestNG

Testing Types: API Testing • UI / E2E • Regression • Reliability • OCR Validation • Template Matching • Visual Regression
Computer Vision / AI: OpenCV (Template Matching) • Tesseract OCR • Entropy Image Quality • GitHub Copilot

Databases: SQL • MongoDB

CI/CD & DevOps: GitHub Actions • Jenkins • Azure DevOps • Docker • ControlHub Server

Cloud & Tools: AWS S3 • AWS EC2 • Postman • Swagger / OpenAPI • Jira • Git / GitHub / Bitbucket

Full Stack (MERN): React.js • Node.js • Express.js • MongoDB

PROFESSIONAL EXPERIENCE

Automation Test Engineer | **Randstad Enterprise Pvt Ltd (Client: Philips Healthcare)**

Jun 2025 – Jun 2026

Bengaluru • Healthcare / Medical Imaging • IVUS Clinical Imaging (IEC 62304 Regulated)

AUTOMATION & FRAMEWORK DESIGN

- Framework Architecture:** Built scalable E2E automation using Python + Behave BDD for Philips IVUS clinical imaging in a regulated IEC 62304 environment.
- UI Automation:** Engineered workflows using SendKey API + iMonkey device integration to accurately simulate real-user behavior in complex medical application scenarios.
- AI-Assisted Development:** Leveraged GitHub Copilot to accelerate script authoring, debugging, and peer code review cycles.

VALIDATION & QUALITY ENGINEERING

- Computer Vision:** Implemented OpenCV template matching for dynamic UI element identification and image-driven visual validation on adaptive medical interfaces.
- OCR Validation:** Applied Tesseract OCR to verify clinical reports, annotations, and on-screen data — ensuring data integrity in regulated healthcare workflows.
- Image Quality Assurance:** Introduced entropy-based image quality validation to ensure diagnostic-level accuracy of IVUS medical imaging outputs.
- API + Visual Regression:** Developed API-driven screenshot capture pipelines to enforce UI consistency and visual regression detection across all build iterations.

CI/CD & DELIVERY

- CI/CD Pipelines:** Configured and maintained GitHub Actions workflows; orchestrated execution via ControlHub server for centralized, scalable test management.
- Reliability Testing:** Conducted rigorous reliability, stability, and regression testing for critical clinical usage; led root-cause analysis and defect triage with dev teams.

Key Achievements

- **Named key contributor** in Philips Rel26H2 reliability testing initiative — life-critical IVUS imaging validated at enterprise scale.

- ▶ Expanded automation coverage via OCR, API, and image-based verification, **significantly reducing manual effort** in a regulated clinical environment.
- ▶ Delivered quality assurance for IEC 62304 regulated software — equivalent to senior QA standards in medical device engineering.

Test Analyst | Art Technology and Software Pvt Ltd

Aug 2022 – Jul 2024

Kochi • Banking & Financial Services • Payment Gateway & Data Lake (Thredd)

AUTOMATION & API TESTING

- **Automation Scripts:** Designed automation scripts using **Python, C#, and SpecFlow (BDD)** with Selenium WebDriver for regression testing of the payment gateway and data lake (Thredd) platform.
- **BDD with SpecFlow:** Implemented **SpecFlow (C#)** BDD framework to translate business requirements into executable Gherkin scenarios, bridging technical and business stakeholders.
- **API & Transaction Validation:** Executed data flow validation scripts for chargeback, authorization, reporting, and reconciliation modules.

DATA & CLOUD VALIDATION

- **Database Testing:** Performed advanced SQL and MongoDB validation ensuring consistency and reliability of critical financial data across distributed systems.
- **AWS S3 Automation:** Implemented data lake validation frameworks using **Python and C# with SpecFlow** to automate large-scale financial dataset integrity checks.
- **CI/CD Integration:** Integrated test suites with Jenkins pipelines, enabling continuous testing and faster deployment feedback loops.

Key Achievements

- ▶ **Reduced AWS S3 validation effort by 40%** through automation, improving data integrity assurance across the data lake platform.
- ▶ Contributed to **stable, on-time releases** across payment processing and data lake modules through optimised automated regression workflows.

FULL STACK DEVELOPMENT (MERN)

MERN Stack Certified Developer — Full product lifecycle capability: React.js frontends, Node.js/Express.js REST APIs, and MongoDB data layers. This cross-functional perspective strengthens **test architecture decisions**, accelerates **dev-QA collaboration**, and reduces defect cycle time through deeper codebase familiarity.

- **Frontend:** React.js — component-based UIs, state management, REST API integration
- **Backend:** Node.js + Express.js — RESTful APIs, middleware, authentication
- **Testing Alignment:** Full stack awareness enables sharper test coverage decisions and faster root-cause analysis in cross-team environments.

EDUCATION & CERTIFICATIONS

B.E. – Mechanical Engineering • Dhanalakshmi Srinivasan Engineering College

2020

- **Python Programming Certification** – GUVI (ID: 069B097t163g23MD5E)
- **Selenium Automation with C# Certification** – GUVI (ID: 069B097t163g23MD5E)
- **MERN Full Stack Development Certification** – GUVI (ID: 483cLnatC287072Ff1)

CURRENTLY LEARNING

- **Playwright** – Modern E2E web automation (TypeScript / Python) for cross-browser testing
- **Advanced API Testing** – Contract testing, performance benchmarking, and security validation patterns

Preferred Locations: Chennai • Bengaluru • Remote | Notice Period: Immediate Joiner